

Exercise 5

Advanced SQL: NULL Functions

TABLE 1: Employees

Employee_id	name	department	Salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL
5			

Q1 SELECT employee_id,
name
COALESCE (Salary, 0) AS salary-with-default
FROM employees;

Employee_id	name	salary-with-default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2 SELECT employee_id,
name
COALESCE (department, not Assigned) AS department_name
FROM employees.

Employee_id	name	department
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

TABLE 2: Orders

Order_id	customer_id	delivery_date
1	201	2024-12-01
2	202	NULL
3	NULL	2024-12-08

Q3. SELECT order_id,
customer_id
FROM Orders
WHERE ISNULL (customer_id, 0) = 0;

Order_id	customer_id
103	NULL

Q4. SELECT order_id,
customer_id
COALESCE (delivery_date, 'Pending') AS delivery_status
FROM Orders

TABLE 3 Students

student_id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Q5. SELECT student_id,
name,
COALESCE (grade, 0) AS Final_grade
FROM Students

Student_id	name	Final_grade
1	Ethan	85
2	Maya	0
3	Olivia	50

Q6. SELECT

COUNT(*) AS not_graded_count
FROM Students
WHERE grade IS NULL,

NOT graded_count
1

TABLE 4: Products

product_id	name	price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Q7 SELECT product_id,
name,
~~price~~

COALESCE (discount, 0) AS final_price
FROM Products

Product_id	name	Final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

TABLE 5: Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	Joseph@mail.com
3	Mia	NULL

Q8. SELECT COUNT (*) AS missing_email_count
FROM customers
WHERE ~~email~~ email IS NULL;

Missing_email_count
2

Q9. SELECT customer_id
name,
COALESCE (email, 'No Email') AS email_display
FROM customers

customer_id	name	email_display
1	Linda	No email
2	Joseph	Joseph@mail.com
3	Mia	No email

TABLE 6: Payments

Payment_id	Method	Status
301	Credit	NULL
302	Paypal	Success
303	NULL	Failed

Q10 SELECT Payment_id
 (COALESCE (method, 'unknown') AS method_display
 Status
 FROM Payments;

Payment_id	Method_display	Status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed

TABLE 7: Inventory

Item_id	Item_name	Quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Q11 SELECT item_id
 item_name
 (COALESCE (quantity, 0) AS quantity_checked
 FROM Inventory

Item_id	Item_name	Quantity
1	Pen	0
2	Notebook	150
3	Eraser	0

TABLE 8: Employees - Extra

emp-id	bonus	commission	Q12. SELECT emp-id COALESCE (bonus, commission) AS first-available-reward FROM Employees_extra;
1	NULL	300	
2	100	NULL	
3	NULL	NULL	

emp_id	First_available_reward
1	300
2	100
3	NULL

TABLE 9: Classes

class_id	subject	room
11	Math	NULL
12	Science	lab A
13	English	NULL

Q13 SELECT COUNT(*) AS no_room-count
FROM classes
WHERE room IS NULL;

no_room-count
2

TABLE 10: Attendance

Student_id	date	Status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Q14 SELECT student_id,
date,
COALESCE(status, 'Not Marked') AS attendance_status
FROM Attendance;

Student_id	date	attendance_status
1	2025-04-01	Not marked
2	2025-04-01	Present
3	2025-04-01	Absent

TABLE 11: BANK

account_id	account_type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

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SELECT account_id
       COALESCE (account_type, 'unknown') AS type_display,
       COALESCE (balance, 0) AS balance_checked
FROM Bank_Accounts;

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account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

TABLE 12: PROJECTS

Project_id	title	start_date	end_date
1	Website revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

```

Tab SELECT project_id,
       title,
       COALESCE (start_date, 'TBA') AS start_display
FROM Projects

```

Project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBA
3	Data Migration	TBA

TABLE 13: Reviews

	Review-id	Product-id	comment	rating
1	1	S01	Great product	4
2	2	S02	NULL	NULL
3	3	S03	Works Fine	3

Q17 SELECT review-id,

product-id

COALESCE (comment, 'No comment') AS comment_display,

COALESCE (rating, 0) AS rating_display

FROM Reviews;

Review-id	product-id	comment_display	rating_display
1	S01	Great product	4
2	S02	No comment	0
3	S03	Works Fine	3

TABLE 14: Suppliers

Supplier-id	name	phone	alt-phone
1	Global Goods	NULL	123456789
2	Best Suppliers	987654321	NULL
3	ValueSource	NULL	NULL

Q18 SELECT supplier-id,

name

COALESCE (phone, alt-phone, 'No contact') AS contact-number

FROM Suppliers

Supplier-id	name	contact-number
1	Global Goods	123456789
2	Best suppliers	987654321
3	Value source works	No contact

Q19 TABLE IS: User-settings

user_id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

Q19. SELECT user_id,
 COALESCE (theme, 'Light') AS theme_set,
 COALESCE (language, 'English') AS language_set,
 COALESCE (timezone, 'UTC') AS timezone_set

FROM User_settings;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

TABLE 16: Maintenance

record_id	machine_id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Q20 SELECT record_id,
 machine_id,
 COALESCE (issue, 'Unknown issue') AS issue_log
 COALESCE (technician, 'Not assigned') AS technician_name

FROM Maintenance;

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex